Hierarchical Shell Manifold Theory

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Abstract

Hierarchical Shell Manifold Theory (HSMT) proposes that the physical universe we observe may be one manifestation of a deeper, interconnected hierarchy of dimensional and organizational regimes, referred to as shells. Within this framework, the familiar structures of space, time, matter, and physical interactions are investigated as potentially emergent features of a more comprehensive underlying mathematical organization. Rather than representing conventional spatial layers, shells describe distinct regimes of effective behavior whose interactions may influence the physical phenomena observed within any particular regime.

HSMT develops this possibility through a graded Hilbert-space architecture equipped with sector projections, reference operators, and intergrade couplings. These structures provide a mathematical setting in which different levels of organization can interact while supporting distinct effective descriptions. Projection methods relate the dynamics of selected sectors to the larger system, allowing effective interactions, spectral responses, and memory effects to be investigated without assuming that the observed physical description is fundamental.

The framework establishes controlled operator constructions, exact Schur–Feshbach reductions, spectral and dimensional diagnostics, and finite relational models exhibiting autonomous collective behavior. These results demonstrate how mathematically distinct effective regimes can arise within a common underlying architecture. They also provide a foundation for investigating possible connections between hierarchical organization and the phenomena described by quantum mechanics, gravitation, particle physics, and cosmology.

A central challenge is determining why nature would realize one particular physical organization from among the mathematically admissible possibilities. To investigate this question, we introduce the Maximal Collective-Relaxation Selection Principle (MCRSP-1) as a candidate physical-selection mechanism. Within a specified family of weighted interaction networks, this principle uniquely determines the interaction distribution that maximizes the slowest collective relaxation rate, yielding explicit mathematical predictions and experimentally distinguishable configurations.

The resulting theory offers a constructive approach to studying how effective physical descriptions might emerge from hierarchical structure. Its operator framework and conditional mathematical results are established within their stated assumptions, while the identification of a unique physical generator, the emergence of observed spacetime, and the universal applicability of the proposed selection mechanism remain open questions. HSMT is therefore presented as a mathematically developed research framework for investigating the possibility that the apparently distinct structures of physical reality arise from a common hierarchical foundation.

Contents

1 Introduction **5**

1.1 What Is a Shell? 5

1.2 Interactions Between Shells . . . 6
1.3 The Mathematical Architecture . . . 6
1.4 From Underlying Structure to Effective Physical Descriptions . . . 7
1.5 Hierarchy, Dimensionality, and Emergent Geometry . . . 7
1.6 Physical Selection and the Search for Fundamental Dynamics . . . 8
1.7 Purpose and Scope of the Present Work . . . 9

2 Logical Status, Scope, and Interpretive Firewalls 10

3 Minimal HSMT Core 12
3.1 Graded Hilbert structure . . . 12
3.2 Geometric realizations . . . 12
3.3 Dynamical consistency axiom DCP-0 . . . 13

4 Graded Operator Structure and Self-Adjointness 13
4.1 Gaussian coupling realization . . . 14

5 Projection, Effective Operators, and Memory 14
5.1 Time-domain memory . . . 15
5.2 Isolation estimate . . . 16
5.3 Projection is not selection . . . 16

6 Spectral Structure and Inverse Problems 16
6.1 Scalar Green function and self-energy . . . 17
6.2 Application A0: semicircle benchmark . . . 17
6.3 Conditional inverse-square branch . . . 18
6.4 WKB frame-density map . . . 19

7 Dimensional Diagnostics and Mode Counting 19
7.1 Mode counting and MS-C0 . . . 20

8 Dynamics, Measurement, Packets, and Records 21
8.1 POVM structure . . . 21
8.2 Packets and localization . . . 21
8.3 Record persistence . . . 21

9 No-Go and Non-Selection Results 22
9.1 NI-0: monotone-coupling no-go . . . 22
9.2 NI-1F: Gaussian measured-pullback monotonicity . . . 23
9.3 S+C minimal nonfundamental benchmark . . . 23
9.4 B3, B5, and B7 . . . 23
9.5 Transport non-selection . . . 23
9.6 Positivity counterexample . . . 24
9.7 OMP-3F valid stop . . . 24
9.8 BEGC no-smuggling comparison . . . 24

9.9 PB-0: physical projector principled stop . . . 24
9.10 PB0-N1 through PB0-N7 . . . 25
9.11 Global non-selection boundary . . . 25

10 Foundational Closure and HSMT-FM0 **26**

11 Conditional Physical Bridges **26**
11.1 Criterion for reopening derivation after FM0 . . . 28

12 Computational and Hardware Benchmarks **28**
12.1 P3 exact benchmark . . . 28
12.2 Six-real-state S and I realizations . . . 29
12.3 P5 controlled discriminator . . . 29
12.4 EW-0 and implementation status . . . 30
12.5 Evidence ladder . . . 30

13 Falsifiability and Empirical Access **30**

14 Candidate Selection Principle: Maximal Collective Relaxation **32**
14.1 Scientific status and relationship to prior optimization literature . . . 32
14.2 Variational formulation . . . 33
14.3 Exact finite spectra and an independent discriminator . . . 34
14.4 Residual freedom: optimizing a grade block is not selecting physics . . . 34
14.5 Experimental test and falsification protocol . . . 35
14.6 Connection to the physical-selection ledger . . . 35

15 The Open Physical Selector Problem **35**
15.1 Concise physical-completion ledger . . . 36

16 Discussion **36**

17 Conclusion **37**

A Notation, Status Vocabulary, and Closure Terminology **38**

B Operator Domains and Self-Adjointness Details **38**

C Exact Feshbach Reduction and Memory Kernel **38**

D Spectral Inversion and Conditional Asymptotics **39**
D.1 Stieltjes consistency checks . . . 39
D.2 A0 noise response . . . 39
D.3 Historical chain . . . 39

E Dimensional Diagnostics, SRB Counts, and MS-C0 **39**

F No-Go and Boundary Ledger **39**

G POVM, Packets, and Records **40**

H P3, P5, S+C, and Spectral-Law Benchmarks **41**
H.1 P3 characteristic polynomial 41
H.2 General power-law discriminator 41
H.3 S+C 41

I Hardware Architecture and Execution Status **41**

J Frozen Checkpoint Ledger **42**

K Conditional Bridges, MC-1, and External Reopening **42**

L Superseded Developmental Material and Provenance **42**

Summary Status Tables **43**

Acknowledgments **43**

1 Introduction

Modern physics describes the universe through several remarkably successful theoretical frameworks. Quantum mechanics governs the behavior of matter and energy at microscopic scales. General relativity describes gravitation and the large-scale geometry of spacetime. The Standard Model characterizes the known elementary particles and their nongravitational interactions, while cosmology investigates the evolution and large-scale structure of the observable universe.

Although these theories have achieved extraordinary experimental success, the relationships between their underlying mathematical structures remain incompletely understood. In particular, quantum theory and gravitation employ fundamentally different descriptions of physical reality, and the origin of the observed dimensionality, particle content, and dynamical structure of the universe remains an open subject of foundational investigation.

Hierarchical Shell Manifold Theory (HSMT) approaches these questions from a different starting point.

Instead of assuming that the familiar organization of physical reality is necessarily fundamental, HSMT investigates whether the universe we observe could represent one effective level within a deeper hierarchy of interconnected physical organization.

The central proposal is that reality may possess multiple regimes of structure, referred to as *shells*, which are mathematically distinguishable but potentially interconnected. Each shell represents a possible regime of effective organization whose observable properties may depend both on its internal structure and on its interactions with other shells.

In this picture, familiar physical phenomena might arise not solely from the properties of one isolated regime, but also from relationships within the larger hierarchical system.

The purpose of HSMT is to develop a mathematical framework in which this possibility can be investigated systematically.

1.1 What Is a Shell?

The term *shell* requires clarification, because it does not refer to an ordinary spherical surface or a layer surrounding the universe in familiar three-dimensional space.

An HSMT shell is a distinct level or regime of mathematical organization within a hierarchical structure.

Different shells may support different effective descriptions, interaction patterns, spectral characteristics, or dimensional behavior. Their distinction is therefore organizational rather than necessarily spatial.

One useful analogy is the relationship between microscopic and macroscopic descriptions of matter.

A material composed of atoms can exhibit collective properties, such as elasticity, temperature, and fluid behavior, that are not naturally expressed in the same language used to describe its individual constituents.

The macroscopic description does not replace the microscopic one. Instead, it captures a different level of organization.

HSMT explores a broader mathematical possibility: that apparently distinct physical descriptions may similarly arise from different regimes within a common underlying structure.

Unlike a conventional hierarchy based only on physical size, however, HSMT allows the organization of these regimes to be associated with different spectral, geometric, and potentially dimensional characteristics.

The integer label assigned to a shell identifies its position within the mathematical hierarchy. It does not, by definition, specify the number of spatial dimensions in that regime.

This distinction permits effective dimensionality to be investigated as a property of the system rather than inserted through the shell label itself.

1.2 Interactions Between Shells

A defining feature of HSMT is that the different regimes of organization need not be dynamically isolated.

The framework permits interactions between shells through mathematical operators that connect their associated state spaces.

These interactions are referred to as *intergrade couplings*.

Their role is conceptually important.

If a selected shell interacts with other sectors of a larger system, its observable behavior may reflect more than its internal dynamics.

Interactions with complementary sectors can modify its effective response, introduce additional spectral structure, and generate memory effects.

Consequently, an observer restricted to one effective regime could encounter behavior whose mathematical origin lies partly in degrees of freedom outside that regime.

This possibility provides the central motivation for exploring hierarchical relationships in fundamental physics.

For example, HSMT investigates whether features associated with quantum behavior, effective interactions, or geometric response might ultimately be understood through relationships between different organizational regimes.

Such connections are research hypotheses, not identifications established merely by introducing a hierarchy.

The mathematical task is to determine what interactions of this kind actually produce and whether their consequences can be connected to observed physics.

1.3 The Mathematical Architecture

To study this proposal, HSMT represents the underlying hierarchy using a graded Hilbert space.

Let $\mathcal{H}$ denote the full state space of the mathematical system.

Its organization into distinct sectors is expressed through the decomposition

$$\mathcal{H} = \bigoplus_{N \in \mathcal{I}} \mathcal{H}_N, \tag{1}$$

where $\mathcal{H}_N$ denotes the state space associated with grade N, and $\mathcal{I}$ is the set of grade labels.

Each sector is identified by an orthogonal projection operator P_N.

The complete family satisfies

$$P_N P_M = \delta_{NM} P_N, \qquad \sum_{N \in \mathcal{I}} P_N = I, \tag{2}$$

where the sum converges in the strong operator topology.

A densely defined reference operator D supplies the additional structure needed for spectral and operator-theoretic investigation.

The minimal mathematical foundation of HSMT can therefore be represented as

$$\boxed{(\mathcal{H}, \{P_N\}_{N\in\mathcal{I}}, D)\,.} \tag{3}$$

This formulation is deliberately general.

It describes how distinct sectors can be organized within one mathematical system without requiring that they already possess a particular spacetime geometry or physical interpretation.

When additional operator assumptions are introduced, the framework permits self-adjoint realizations, intergrade interactions, spectral analysis, and effective-sector reductions.

The resulting mathematical architecture provides a controlled setting for investigating how different levels of organization can coexist and influence one another.

1.4 From Underlying Structure to Effective Physical Descriptions

One of the most important questions motivating HSMT is how a particular level of organization can possess an effective description of its own while remaining part of a larger system.

This question can be investigated through projection methods.

Suppose a mathematical sector is selected by an orthogonal projection P, with complementary projection

$$Q = I - P. \tag{4}$$

For a suitable self-adjoint operator H, the complete system can be decomposed into interactions within the selected sector, interactions within its complement, and couplings between them.

The Schur–Feshbach formalism then provides an exact relationship between the full operator and the effective response of the selected sector.

In resolvent form, the effective operator contains the contribution

$$\Sigma_P(z) = H_{PQ}(z - H_{QQ})^{-1}H_{QP}, \tag{5}$$

where $\Sigma_P(z)$ represents the influence of the complementary sector on the selected one.

This expression captures an essential feature of HSMT.

The mathematical behavior associated with one regime can depend on its coupling to a larger structure, even when those additional degrees of freedom do not appear explicitly in the reduced description.

The framework therefore allows effective dynamics to be studied as consequences of the organization and interactions of the complete system.

This does not require assuming that every observed physical interaction has already been derived from intergrade coupling.

Rather, it provides a mathematical mechanism through which such possibilities can be investigated.

1.5 Hierarchy, Dimensionality, and Emergent Geometry

A further motivation for HSMT is the possibility that dimensionality and geometric behavior may depend on the scale and manner in which a system is examined.

In ordinary geometric descriptions, dimension is commonly treated as a fixed property of the underlying space.

In more general mathematical settings, however, different notions of effective dimension can be

constructed from scaling, spectral response, diffusion, or resolution-dependent measurements.

HSMT permits such dimensional diagnostics to be associated with its graded sectors.

For a specified scale-dependent quantity $Q_N(\ell)$, one may define the logarithmic scaling exponent

$$d_N(\ell) = \frac{\mathrm{d}\ln Q_N(\ell)}{\mathrm{d}\ln \ell}. \tag{6}$$

The physical interpretation of this exponent depends on the quantity being measured.

Nevertheless, the construction illustrates how effective dimensional behavior can be studied without identifying the grade label itself with spatial dimension.

This opens a mathematical route for investigating hierarchical relationships between different geometric or spectral regimes.

It also highlights an important distinction: obtaining a dimension-like scaling exponent does not, by itself, establish the dimensionality of physical spacetime.

A physical interpretation requires a connection between the mathematical diagnostic and an operationally defined observable.

1.6 Physical Selection and the Search for Fundamental Dynamics

The graded architecture of HSMT admits a variety of mathematically consistent realizations.

This flexibility is useful for constructing and studying different types of hierarchical systems.

However, it raises a fundamental physical question:

> What determines which mathematical organization and dynamical laws are actually realized in nature?

The existence of multiple admissible structures means that mathematical consistency alone is not sufficient to identify a unique physical realization.

A genuine physical-selection principle would need to explain why particular sectors, interaction patterns, and effective dynamics are realized rather than other mathematically possible alternatives.

As one candidate approach, this paper examines the Maximal Collective-Relaxation Selection Principle (MCRSP-1).

The proposal investigates whether adaptive interaction networks with constrained total coupling resources might select configurations that maximize their slowest collective relaxation rate.

Within a specified family of weighted path networks, this optimization leads to a unique coupling distribution

$$w_i^* = \frac{6W}{n(n^2-1)} i(n-i), \tag{7}$$

where n is the number of vertices and W is the total interaction weight.

For a five-site example, the selected coupling ratio is

$$w_1 : w_2 : w_3 : w_4 = 2 : 3 : 3 : 2. \tag{8}$$

This result demonstrates that a mathematically specified selection principle can determine nontrivial interaction structures in a restricted class of models.

It does not yet establish that nature obeys this principle or that it selects the complete fundamental dynamics required by HSMT.

Its value lies in providing a concrete and testable candidate for further investigation.

1.7 Purpose and Scope of the Present Work

The purpose of this paper is to present Hierarchical Shell Manifold Theory as a coherent mathematical framework for investigating the relationship between hierarchical organization and effective physical descriptions.

The analysis develops the graded operator architecture, examines controlled intergrade interactions, and establishes exact relationships between selected sectors and their complementary degrees of freedom.

It also develops spectral and dimensional diagnostics, measurement constructions, relational models, and finite benchmark systems through which particular features of the framework can be examined.

These mathematical results provide a structured foundation for exploring the possibility that different effective descriptions of physical reality may be connected through a common hierarchical organization.

The broader scientific motivation is to investigate whether phenomena currently described by separate physical theories may share deeper structural relationships.

At its present stage, HSMT does not claim a completed derivation of quantum mechanics, general relativity, the Standard Model, or cosmological evolution from one independently established fundamental law.

Instead, it provides a mathematical framework, a collection of rigorous and conditional results, and a developing program of physical-selection and experimental tests.

The following sections present this framework in detail, beginning with its operator structure and progressing through effective dynamics, spectral behavior, dimensional diagnostics, relational constructions, and candidate physical selection.

The central objective is to determine how far the mathematics of hierarchical organization can take us toward understanding the possible emergence of familiar physical reality from a deeper common foundation. The minimal frozen mathematical data are

$$(\mathcal{H}, \{P_N\}, D), \tag{9}$$

where $\mathcal{H}$ is a Hilbert-state arena, the P_N are mutually orthogonal grade projections, and D is a densely defined reference operator. A geometric realization may instead be denoted schematically by

$$\mathcal{M} = \bigsqcup_N \mathcal{M}_N, \tag{10}$$

but the disjoint-union notation does not turn the pieces into physical spacetime manifolds by definition. These ingredients are sufficient to organize a useful analysis; specific dynamical and geometric conclusions arise only under additional, declared hypotheses.

From grades to effective descriptions. In this picture, a single operator may have diagonal blocks describing activity within grades and off-diagonal blocks transmitting influence between them. Resolvent reduction and its time-domain counterpart make that relation exact. Spectral measures record the response of a selected sector to the surrounding degrees of freedom. Scaling observables allow one to ask whether different organizational regimes exhibit different effective dimensional behavior. Relational graph constructions and programmable finite systems provide examples in which the formalism can be calculated and checked explicitly.

The physical-selection question. The presence of multiple mathematically consistent effective descriptions raises a question that the architecture alone cannot answer: which operator, sector, and interpretation are realized in nature? We examine this directly rather than concealing the missing choice inside a convenient kernel, a projector, or a dimensional label. A particular proposal, MCRSP-1, asks whether adaptation toward fast collective relaxation could select the strengths of intersite interactions. Its value is that it yields definite mathematical outcomes and discriminating tests, not that it has already been confirmed as a universal law.

Contributions and organization. This article first builds the graded architecture and its effective-dynamics machinery; then it explains spectral, measurement, dimensional, and relational constructions; next it collects the precise boundaries of those results and their experimental realizations; and finally it develops and tests the candidate selection principle. We preserve the proofs, counterexamples, and declared domains of validity throughout. The aim is to give readers both the constructive framework and an accurate account of what must still be supplied to turn it into a physical theory.

2 Logical Status, Scope, and Interpretive Firewalls

A theory of emergence must tell a reader when it is describing an exact mathematical consequence and when it is suggesting a physical interpretation. For example, a graph may have a well-defined notion of distance even though its vertices are not points in space; a matrix may generate oscillations even though it has not been identified with a physical Hamiltonian. HSMT uses a simple claim-status system so that its mathematical constructions can be read positively, without mistaking them for experimentally established physics. The classifications below are tools for clarity, not replacements for the central ideas.

The distinction between mathematical derivation and physical interpretation is sufficiently central that claim status is made explicit throughout the paper.

Definition 2.1 (Claim-status vocabulary)**.** *A statement is labeled according to one of the following classes.*

AXIOM *An adopted primitive of the framework.*

DEFINITION
A mathematical construction or terminological convention.

THEOREM *A rigorous result under explicit hypotheses.*

DERIVED RESULT
A consequence of stated assumptions inside a declared model class.

NO-GO *A theorem, inequality, or counterexample excluding a stronger conclusion within a stated class.*

NON-SELECTION
The present premises do not determine a desired structure; impossibility under future extensions is not claimed.

VALID STOP
A principled boundary at which neither a proposition nor its negation has been derived.

CONDITIONAL RESULT

A conclusion that follows only after an additional bridge or realization assumption is supplied.

HYPOTHESIS

A motivated but presently unproved proposal.

BENCHMARK

A mathematical, numerical, emulator, circuit-model, or experimental test of a declared realization.

OPEN PROBLEM

A question left unresolved by the present framework.

The central interpretive firewalls are summarized in Table 1.

Table 1: Interpretive firewalls of mature HSMT.

Mathematical structure	**Permitted statement**	**Not derived automatically**
Grade N	Hierarchy/sector label	Physical dimension
Evolution parameter τ	Mathematical ordering parameter	Operational physical time
Generator H or D	Evolution/spectral operator	Physical clock or temporal metric
Spectral/scaling exponent	Diagnostic of a specified quantity	Observed spatial dimension
Mode count	Combinatorial scaling	Physical dimension
Propagation structure	Mathematical characteristic information	Lorentzian causal cone
Projected sector P	Reduced mathematical sector	Uniquely realized physical world
POVM outcome	Measurement label	Physical position
Record observable	Conserved information	Arrow of time or irreversible measurement

The methodological rule may be stated compactly:

> *A mathematical object is not considered a derived physical observable merely because it can be assigned the name of one.*

Three kinds of negative conclusion are also distinguished. A *no-go* result rules out a stronger statement within a specified class. A *non-selection* result establishes only that current premises do not determine the desired structure. A *valid stop* records a principled unresolved endpoint. These categories will be used explicitly in Section 9.

Finally, several notions of closure are kept separate:

- *Foundational closure*: the minimal architecture has been reduced to a frozen core.
- *Branch closure*: a particular derivation branch has reached its legitimate endpoint, positive, negative, partial, or unresolved.
- *Valid stop*: no licensed continuation exists from the present premises, but neither side of the unresolved proposition has been proved.

- *Internal derivational closure*: the presently identified branches from the current premise set have been exhausted to their logical boundary.
- *Physical closure*: the missing physical selector and operational bridge have been supplied. This has *not* been achieved.

3 Minimal HSMT Core

The idea of a hierarchy becomes mathematically useful once it is possible to say what counts as a level, what states belong to that level, and how levels exchange influence. HSMT addresses these questions using three ingredients: a common state space, projections that divide it into sectors, and a reference operator. Everything that follows can be understood as exploring what these ingredients permit and what additional information would be needed to identify them with nature.

The underlying idea of a graded system can be expressed without specifying a geometry or preferred dynamics. This section explains the basic objects, how grades partition the state space, and what additional assumptions are needed before an operator generates a particular evolution.

3.1 Graded Hilbert structure

Imagine describing a large system while keeping track of which organizational level a state belongs to. An orthogonal projection is a precise mathematical way to perform that bookkeeping: it extracts the part of a state assigned to one level. The collection of such projections makes the hierarchy explicit without presupposing that its levels are literal spatial dimensions.

Let $\mathcal{H}$ be a complex separable Hilbert space and let

$$\{P_N\}_{N\in\mathcal{I}} \tag{11}$$

be a countable family of mutually orthogonal projections such that

$$P_N P_M = \delta_{NM} P_N, \qquad \sum_{N\in\mathcal{I}} P_N = I \tag{12}$$

in the strong operator topology on the sector under consideration. Define

$$\mathcal{H}_N = P_N\mathcal{H}, \qquad \mathcal{H} = \bigoplus_{N\in\mathcal{I}} \mathcal{H}_N. \tag{13}$$

The label N is the *grade*. Equation (13) is structural; it contains no assertion that N is physical dimension.

Let D be a densely defined reference operator on $\mathcal{H}$. The minimal frozen data are therefore the triplet (9). Depending on realization, D may enter spectral analysis, effective reduction, or a closed evolution law. Its physical interpretation is not fixed at the foundational level.

3.2 Geometric realizations

The word *manifold* in the name points toward possible geometric interpretations of the shell structure. One may attach geometric spaces to grades, but geometry enters as a realization of the algebraic structure rather than as a fact built into the definition of a grade. This distinction leaves room to investigate geometry emerging from relations among states.

A geometric realization may attach to the grade structure a disjoint union such as (10). In that case $\mathcal{M}_N$ is a grade-associated geometric object. The mature formulation does not infer that each $\mathcal{M}_N$ is an observed spacetime manifold or that the index N measures physical dimension.

3.3 Dynamical consistency axiom DCP-0

A hierarchy alone does not move. To study change one must specify a rule for how states evolve. The minimal closed-system requirement used in some HSMT realizations is preservation of a positive quadratic norm: the total mathematical state is not amplified or destroyed by its autonomous evolution. This supplies a disciplined starting point for dynamics while leaving the interpretation of an evolution parameter to physical calibration.

Where a closed evolution is introduced, HSMT adopts the minimal dynamical axiom:

DCP-0. Closed evolution is norm preserving.

In a self-adjoint generator realization one may write

$$U(\tau) = e^{-\mathrm{i}H\tau}, \qquad \|U(\tau)\psi\| = \|\psi\|. \tag{14}$$

The parameter τ is deliberately distinguished from physical time. DCP-0 does not by itself derive an operational clock, Lorentzian signature, causal structure, quantum ontology, or an arrow of time.

4 Graded Operator Structure and Self-Adjointness

After defining levels, the next question is how they influence one another. An operator can act within a single level, but it can also connect one level to another. The diagonal parts describe activity internal to a shell; the off-diagonal parts describe intergrade communication. With suitable conditions these pieces can be combined into a mathematically controlled operator, even when the internal dynamics are infinite-dimensional.

An effective hierarchy needs a mathematically controlled way for sectors to interact. Grade-diagonal operators describe within-sector behavior; intergrade terms describe communication between sectors. Self-adjointness gives a rigorous setting for some of these constructions and ensures that the operator calculations are not based merely on formal matrices.

For any operator A whose grade blocks are defined, write

$$A_{NM} = P_N A P_M. \tag{15}$$

Formally,

$$A = \sum_{N,M} A_{NM} = A_{\mathrm{diag}} + A_{\mathrm{hop}}, \tag{16}$$

with

$$A_{\mathrm{diag}} = \sum_{N} A_{NN}, \qquad A_{\mathrm{hop}} = \sum_{N \neq M} A_{NM}. \tag{17}$$

Nearest-neighbor hopping is an additional model restriction,

$$A_{NM} = 0 \quad \text{for} \quad |N - M| > 1, \tag{18}$$

not a consequence of the abstract grade structure.

Theorem 4.1 (Bounded intergrade perturbation)**.** *Let D_0 be self-adjoint on* $\mathrm{Dom}(D_0) \subset \mathcal{H}$ *and let T be a bounded self-adjoint operator on $\mathcal{H}$. Then*

$$D = D_0 + T \tag{19}$$

is self-adjoint on $\mathrm{Dom}(D_0)$*.*

Proof. A bounded self-adjoint T is D_0-bounded with relative bound zero. The Kato–Rellich theorem therefore gives self-adjointness of $D_0 + T$ on $\mathrm{Dom}(D_0)$. □

The theorem is insensitive to whether T is diagonal or purely intergrade. For a nearest-neighbor realization one may write schematically

$$T = \sum_N (T_{N,N+1} + T_{N+1,N}). \tag{20}$$

The theorem establishes operator control for the declared realization; it does not establish a uniquely selected universal physical "Master Operator."

4.1 Gaussian coupling realization

One intuitive way to model communication across the hierarchy is to make nearby grades interact more strongly than distant grades. A Gaussian profile provides a smooth example: its width controls how quickly influence falls with grade separation. The usefulness of this profile is computational and structural; the origin of the profile itself remains a separate question.

A historically productive realization uses a localized intergrade weight of Gaussian form,

$$w(N - M) \propto \exp\left[-\frac{(N - M)^2}{2\sigma^2}\right]. \tag{21}$$

Within one frozen numerical construction,

$$\sigma_0 = \frac{\sqrt{2}}{4} \tag{22}$$

produced nearest-neighbor dominance. This is retained as a numerical/model-specific result. Neither the Gaussian functional form nor the value (22) is selected by the minimal premise set. Possible generalizations such as $\sigma \mapsto \sigma_N$ or grade-dependent separations remain model hypotheses unless supplied by a new derivation or empirical bridge.

5 Projection, Effective Operators, and Memory

How can a subsystem behave as though it has laws of its own while still being coupled to a larger system? This question lies at the center of HSMT. Projection keeps track of the sector one wishes to describe, while exact operator reduction records the influence of everything left out. The resulting effective description need not be a simple copy of the original: it may acquire energy dependence and, in a time-dependent representation, memory.

Projection answers a constructive question: if attention is restricted to one sector, what influence does the rest of the system exert on it? The Schur–Feshbach identity packages that influence into a resolvent-dependent self-energy. In time-dependent realizations, the same elimination leads naturally to memory of past interactions.

Let P be an orthogonal projection onto a selected mathematical sector and let

$$Q = I - P. \tag{23}$$

For a self-adjoint operator H, define the block operators

$$H_{PP} = PHP, \quad H_{PQ} = PHQ, \quad H_{QP} = QHP, \quad H_{QQ} = QHQ. \tag{24}$$

The exact reduction is standard Schur–Feshbach theory.

Theorem 5.1 (Exact projected resolvent)**.** *Let z be such that $z - H_{QQ}$ is invertible on $Q\mathcal{H}$. Define*

$$F_P(z) = z - H_{PP} - H_{PQ}(z - H_{QQ})^{-1}H_{QP}. \tag{25}$$

Whenever $F_P(z)$ is invertible on $P\mathcal{H}$,

$$P(z - H)^{-1}P = F_P(z)^{-1}. \tag{26}$$

The corresponding exact self-energy is

$$\Sigma_P(z) = H_{PQ}(z - H_{QQ})^{-1}H_{QP}, \tag{27}$$

and one may write

$$H_{\text{eff}}(z) = H_{PP} + \Sigma_P(z). \tag{28}$$

The correction (27) is exact under the resolvent hypotheses; it is not a perturbative ansatz.

5.1 Time-domain memory

An observer who follows only part of a system may see effects of earlier states because the unobserved degrees of freedom continue to respond and feed information back. In the projected equations, this appears not as an arbitrary correction but as an integral over prior evolution. The memory kernel is the mathematical record of the omitted sector's influence.

For a declared evolution law

$$\mathrm{i}\frac{\mathrm{d}}{\mathrm{d}\tau}\left|\Psi(\tau)\right\rangle = H\left|\Psi(\tau)\right\rangle, \tag{29}$$

write $\psi_P = P\Psi$ and $\psi_Q = Q\Psi$. Solving the Q equation formally gives

$$\psi_Q(\tau) = e^{-\mathrm{i}H_{QQ}(\tau-\tau_0)}\psi_Q(\tau_0) \tag{30}$$

$$- \mathrm{i}\int_{\tau_0}^{\tau} e^{-\mathrm{i}H_{QQ}(\tau-s)}H_{QP}\psi_P(s)\,\mathrm{d}s. \tag{31}$$

Substitution yields

$$\mathrm{i}\frac{\mathrm{d}\psi_P}{\mathrm{d}\tau} = H_{PP}\psi_P(\tau) + H_{PQ}e^{-\mathrm{i}H_{QQ}(\tau-\tau_0)}\psi_Q(\tau_0) \tag{32}$$

$$- \mathrm{i}\int_{\tau_0}^{\tau} K(\tau - s)\psi_P(s)\,\mathrm{d}s, \tag{33}$$

with exact memory kernel

$$K(\Delta\tau) = H_{PQ}e^{-\mathrm{i}H_{QQ}\Delta\tau}H_{QP}. \tag{34}$$

Memory here is a consequence of eliminating Q degrees of freedom. It is not an independent microscopic

arrow of time.

5.2 Isolation estimate

An effective level becomes more autonomous when its coupling to other levels is weak or when the surrounding states are sufficiently far from the energies being examined. Operator estimates make this intuition quantitative and identify regimes in which a sector can be studied approximately on its own.

If $H_{QP} = H_{PQ}^{\dagger}$ and z lies away from $\mathrm{spec}(H_{QQ})$, then

$$\|\Sigma_P(z)\| \le \|H_{PQ}\|^2 \, \|(z - H_{QQ})^{-1}\| \le \frac{\|H_{PQ}\|^2}{\mathrm{dist}(z, \mathrm{spec}(H_{QQ}))}. \tag{35}$$

This provides a transparent weak-coupling/isolation criterion.

5.3 Projection is not selection

Projection explains how a chosen part of a system behaves. It is a powerful descriptive operation, but the choice of which part to observe must come from elsewhere. In physical applications that choice may be established by preparation, measurement, stability, or an independent law. HSMT keeps this distinction visible so its reductions can be used without presuming the origin of the observed sector.

The logical input to Theorem 5.1 includes P. The theorem does not derive P. This distinction is later formalized by PB-0: Feshbach reduction is conditional dynamics given a decomposition, not a selector.

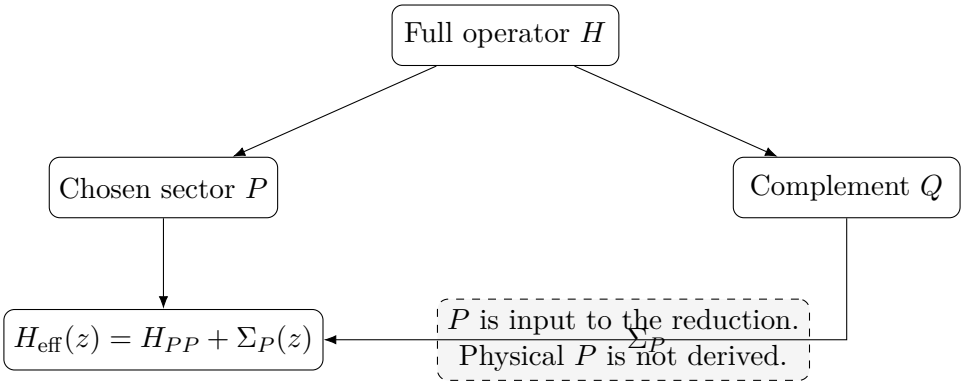


Figure 1: Projection does not equal physical selection. The Schur–Feshbach map exactly reduces a chosen sector; it does not choose that sector.

6 Spectral Structure and Inverse Problems

A system's spectrum tells us how it responds. In HSMT, the spectral viewpoint provides a practical bridge between the internal organization of a hierarchy and the behavior an effective observer might detect. Rather than trying to inspect every hidden degree of freedom, one can ask what their combined influence leaves in an accessible response function.

The spectrum is a way of recording how a system responds. In HSMT, spectral data can be used in two directions: a chosen interaction model predicts a response, and an observed or simulated response may constrain an unknown contact measure or coupling. Inverse problems connect the operator framework to quantities that can be compared and reconstructed.

6.1 Scalar Green function and self-energy

A Green function is a compact way of describing how a system reacts to disturbances. The self-energy measures the change in that response caused by interactions with degrees of freedom outside a chosen sector. This lets the framework translate an otherwise complicated environment into a measurable or reconstructible influence on effective dynamics.

In the scalar realization used in Application A0, the visible resolvent is written

$$G(z) = \frac{1}{z - \omega_0 - \Sigma(z)}. \tag{36}$$

Whenever $G(z) \neq 0$,

$$\Sigma(z) = z - \omega_0 - G(z)^{-1}. \tag{37}$$

The map (37) is algebraically exact within the declared scalar model.

Let μ be a positive contact spectral measure and define

$$m(z) = \int \frac{\mathrm{d}\mu(M)}{z - M}, \qquad \Sigma(z) = g^2 m(z). \tag{38}$$

For $\operatorname{Im} z > 0$, $\operatorname{Im} m(z) \leq 0$. If $\mathrm{d}\mu(M) = \rho(M)\mathrm{d}M$ locally, then

$$\rho(E) = -\frac{1}{\pi g^2} \lim_{\eta \to 0^+} \operatorname{Im} \Sigma(E + \mathrm{i}\eta). \tag{39}$$

At finite resolution,

$$\rho_\eta(E) = -\frac{1}{\pi g^2} \operatorname{Im} \Sigma(E + \mathrm{i}\eta), \tag{40}$$

which is the Poisson-broadened reconstruction of the underlying density.

6.2 Application A0: semicircle benchmark

Before interpreting a reconstructed response as physics, the mathematical machinery must work on examples whose answers are already known. Application A0 serves this purpose: the chosen contact measure has an analytic form, making it possible to compare numerical inversion against an exact reference. Its role is to validate an analysis method, not to claim that nature uses this measure.

The frozen benchmark uses

$$\omega_0 = 0.3, \qquad g^2 = 0.4, \qquad \eta = 0.02, \tag{41}$$

with normalized semicircle contact density

$$\rho_{\mathrm{sc}}(M) = \begin{cases} \dfrac{\sqrt{4 - M^2}}{2\pi}, & |M| \leq 2, \\ 0, & |M| > 2. \end{cases} \tag{42}$$

Its Stieltjes transform is

$$m_{\mathrm{sc}}(z) = \frac{z - \sqrt{z^2 - 4}}{2}, \tag{43}$$

where the branch is chosen so that $\sqrt{z^2-4} \sim z$ as $|z| \to \infty$. For $-2 < E < 2$,

$$m_{\mathrm{sc}}(E + \mathrm{i}0) = \frac{E - \mathrm{i}\sqrt{4 - E^2}}{2}, \tag{44}$$

which recovers (42) through (39).

The corresponding memory transform is

$$\int_{-2}^{2} \rho_{\mathrm{sc}}(M) e^{-\mathrm{i}M\tau} \, \mathrm{d}M = \frac{J_1(2\tau)}{\tau}. \tag{45}$$

Since $J_1(2\tau) \sim (\pi\tau)^{-1/2} \cos(2\tau - 3\pi/4)$ up to subleading corrections,

$$\mathrm{memory}(\tau) \sim \tau^{-3/2} \times \mathrm{oscillation}, \tag{46}$$

so the exact asymptotic exponent is $p = 3/2$.

The completed numerical benchmark produced the results in Table 2.

Table 2: Frozen Application A0 benchmark results.

Quantity	**Frozen result**
Noiseless $G \to \Sigma$ maximum absolute error	$\approx 6.66 \times 10^{-16}$
Noiseless $G \to \Sigma$ relative L^2 error	$\approx 3.73 \times 10^{-16}$
Finite-resolution interior density NRMSE	$\approx 0.984\%$
Fitted memory exponent	$p = 1.49805$ versus exact 1.5
Numerical total measure weight	≈ 0.9999992
Off-support poles in declared scan $[-5, 5]$	none found for the declared parameter set
10^{-4} multiplicative complex G noise	Σ relative L^2 error $\approx 3.77 \times 10^{-4}$
10^{-3} multiplicative complex G noise	Σ relative L^2 error $\approx 3.65 \times 10^{-3}$
10^{-2} multiplicative complex G noise	Σ relative L^2 error $\approx 3.62 \times 10^{-2}$

These results validate the numerical inverse engine against the declared analytic benchmark; they do not establish that the semicircle contact measure is physically realized.

6.3 Conditional inverse-square branch

Some candidate spectral behaviors have particularly simple mathematical signatures. One investigated branch leads to an inverse-square operator and a characteristic boundary index. Examining that chain is useful because each exponent can be traced to a declared premise, showing exactly what follows when that spectral ansatz is adopted.

A historical asymptotic branch uses an exponent β_Ω taking the values

$$\beta_\Omega = \frac{5}{6} \qquad \text{or} \qquad \beta_\Omega = \frac{7}{6}. \tag{47}$$

The inherited coefficient map is

$$c(\beta_\Omega) = \frac{\beta_\Omega(2 - \beta_\Omega)}{4}, \tag{48}$$

so both branches give

$$c = \frac{35}{144}, \qquad -\frac{c}{y^2} = -\frac{35}{144y^2}. \tag{49}$$

For the corresponding inverse-square equation, the Bessel-type index is

$$\nu_B = \sqrt{\frac{1}{4} - c} = \frac{1}{12}. \tag{50}$$

This value is mathematically correct within the historical inverse-square branch; it is not independently selected by the minimal HSMT core.

Under the additional historical identifications, the retained conditional chain is

$$\begin{aligned}\Omega(y) \sim y^{5/6} \text{ or } y^{7/6} \Rightarrow -\frac{35}{144y^2} \Rightarrow \nu_B = \frac{1}{12} \Rightarrow \mathrm{DtN} \sim A^{1/12} \\ \Rightarrow M^{-1/6} \Rightarrow \mathrm{d}\mu(M) \sim M^{-1/3}\mathrm{d}M \Rightarrow \Sigma \sim p^2 E^{-1/3} \Rightarrow E \sim p^{3/2}.\end{aligned} \tag{51}$$

Equation (51) is retained explicitly as a *conditional historical scaling chain*. The intermediate arrows after the inverse-square step rely on source-specific identifications and are not reconstructed here as new derivations.

6.4 WKB frame-density map

A semiclassical approximation can connect the distribution of modes with the growth of an underlying confining profile. This offers another way to understand where candidate spectral powers come from and also shows why a mathematically allowed exponent need not be a uniquely selected one.

A separate branch begins from a chosen asymptotic superpotential

$$W(y) \sim g y^s, \qquad g > 0. \tag{52}$$

The frozen WKB counting result is

$$\frac{\mathrm{d}n}{\mathrm{d}y} \sim y^s. \tag{53}$$

Defining $\beta_{\mathrm{fr}} = s$, one obtains

$$n(y) \sim C y^{s+1}, \qquad y_n \sim n^{1/(s+1)}. \tag{54}$$

For harmonic confinement, $s = 1$ and therefore

$$y_n \sim n^{1/2}. \tag{55}$$

Proposition 6.1 (WKB density map and non-selection)**.** *Given an inserted asymptotic superpotential $W(y) \sim g y^s$ in the declared WKB regime,* (53)–(54) *follow. The current HSMT premises do not select s or uniquely select W.*

The frame index n is not automatically the grade N. Likewise a coordinate rewrite such as $s = 1/a$ is a reparameterization unless a separate dynamical principle selects a.

7 Dimensional Diagnostics and Mode Counting

What would it mean for a shell to have a dimension? HSMT treats this as a measurement question rather than a property read off from its name. Different tests of effective dimension can follow how

volume, response, resolution, or dynamics change with scale. They may agree in special realizations, but it is important to establish that agreement rather than assume it.

Dimension is treated as a property to be measured, not a label to be assigned. One may probe how counts, returns, volumes, or responses change with scale; different probes need not return the same exponent. This distinction makes running or fractional effective dimension a meaningful question without identifying the hierarchy index with dimension.

Let $Q_N(\ell) > 0$ be a declared scale-dependent quantity associated with grade N. Define

$$d_N(\ell) = \frac{\mathrm{d}\ln Q_N(\ell)}{\mathrm{d}\ln \ell} = \frac{\ell}{Q_N(\ell)}\frac{\mathrm{d}Q_N}{\mathrm{d}\ell}. \tag{56}$$

If $Q_N(\ell) = C\ell^d$ on a scale interval, then $d_N(\ell) = d$ there. If no exact power law holds, d_N is a running logarithmic exponent. A finite-resolution estimator is

$$d_N[\ell_1, \ell_2] = \frac{\ln Q_N(\ell_2) - \ln Q_N(\ell_1)}{\ln \ell_2 - \ln \ell_1}. \tag{57}$$

The mature framework distinguishes

$$d_{\mathrm{geom}}, \qquad d_{\mathrm{res}}, \qquad d_{\mathrm{spec}}, \qquad d_{\mathrm{dyn}}. \tag{58}$$

No equality among the quantities in (58) is assumed or generally derived.

The logical order is therefore

$$N \longrightarrow \text{specify or derive } Q_N(\ell) \longrightarrow d_N(\ell), \tag{59}$$

not $N = d_N$. Neither a monotone progression of d_N across grades nor a fractional interpolation of physical dimension between grades is derived by the current premise set.

7.1 Mode counting and MS-C0

The number of available states can grow with system size in a way reminiscent of geometric dimension. That resemblance can be informative, yet counting independent modes does not on its own tell us what a ruler, clock, or laboratory probe would measure as spatial dimension.

The frozen SRB analysis contains the asymptotic mode counts

$$Q_I(M) \sim \frac{M^3}{6}, \qquad Q_H(M) \sim \frac{M^3}{6} \tag{60}$$

for large M in the declared construction. Therefore

$$\frac{\mathrm{d}\ln Q}{\mathrm{d}\ln M} \longrightarrow 3. \tag{61}$$

No-Go Result 7.1 (MS-C0: mode count does not select physical dimension)**.** *If $Q(M) \sim CM^k$ with $C > 0$, then the logarithmic count exponent tends to k. This establishes scaling with respect to the variable M. It does not establish that physical dimension equals k unless M and Q are independently given the required operational geometric interpretation.*

Accordingly, (60) does not imply three-dimensional physical space. The frozen datum $q_I(4) = 2$ is likewise a structural fact of its combinatorial/operator construction and is not by itself a physical transition or dimension statement.

8 Dynamics, Measurement, Packets, and Records

For a mathematical hierarchy to become relevant to experiment, it must eventually describe not only state evolution but also what can be observed. This section examines the formal ingredients of measurement: probability assignments, localized or approximately localized packets, and records that remain identifiable as a system evolves. Together they show how accessible information can be organized within a graded system.

To interpret a graded model operationally, one must specify what can be prepared and measured and which features persist during evolution. This section describes the mathematical role of measurement operators, localized packets, and conserved records. These constructions make explicit which observables belong to a declared model and which physical interpretations remain to be supplied.

8.1 POVM structure

A positive operator-valued measure (POVM) associates possible measurement outcomes with positive operators. This is a flexible way of describing what a restricted observer can distinguish. HSMT uses such structures to explore the information available within sectors or internal fibres, without assuming that every outcome is a location in ordinary space.

Let X be an outcome space. A positive-operator-valued measure E assigns positive operators $E(\Delta) \geq 0$ to measurable sets $\Delta \subseteq X$, satisfies $E(X) = I$, and is countably additive in the weak operator topology. For a state ρ,

$$p_\rho(\Delta) = \mathrm{Tr}[\rho E(\Delta)]. \tag{62}$$

The frozen PC sequence establishes that the relevant internal-fibre HSMT construction admits a canonical POVM. "Canonical" here is relative to that internal structure; it does not mean unique physical detector, observer, spacetime localization, or measurement ontology.

Informational completeness on a state class $\mathcal{S}$ means that equality of all outcome probabilities implies equality of states:

$$\mathrm{Tr}(\rho E_x) = \mathrm{Tr}(\sigma E_x)\ \forall x \quad \Longrightarrow \quad \rho = \sigma, \qquad \rho, \sigma \in \mathcal{S}. \tag{63}$$

The frozen result is informational completeness on the declared internal fibre. This does not make the outcome labels physical positions.

8.2 Packets and localization

A packet is a state concentrated with respect to some chosen observable or family of probes. Its meaning depends on what counts as localization. In a graded setting one must establish the relevant measurement map before interpreting packet concentration as spatial position.

Packet-like states may be useful representations or probes. The frozen PC sequence does not, however, select a canonical localized packet. Moreover, OMP-3F closes at a valid stop: packet-frame completeness remains undetermined. Neither completeness nor incompleteness is licensed.

The scalar Q_N used in the measurement layer is a diagnostic only. There is no contradiction between a full internal-fibre POVM being informationally complete and Q_N being only one compressed scalar functional of the measurement structure.

8.3 Record persistence

An observation becomes useful when its result can persist long enough to be compared with another. Mathematically, a record may be represented by a quantity preserved under the chosen dynamics. This

supplies a clear test for persistence while leaving the broader questions of irreversibility and the arrow of time for a separate physical account.

Let H be self-adjoint and let Q_a be bounded, or satisfy the appropriate strong-commutation/domain conditions. If

$$[H, Q_a] = 0, \tag{64}$$

then

$$e^{\mathrm{i}H\tau} Q_a e^{-\mathrm{i}H\tau} = Q_a \tag{65}$$

and hence

$$\mathrm{Tr}[\rho(\tau) Q_a] = \mathrm{Tr}[\rho(0) Q_a]. \tag{66}$$

This is the frozen RS-0 persistence condition. Conservation of Q_a does not by itself establish a physical clock, an observer, irreversible measurement, or a thermodynamic arrow.

A possible future temporal bridge would require additional operational structure: a monotonic clock variable, reproducible subsystem comparisons, and an independently justified temporal normalization. Those are not supplied by RS-0 itself.

9 No-Go and Non-Selection Results

A constructive framework becomes stronger when it can distinguish real implications from attractive shortcuts. In HSMT, several plausible ideas were tested directly: whether a monotone coupling could automatically supply locality, whether a projection could determine its own physical sector, or whether a positive relational operator could already be a spacetime metric. The resulting counterexamples map the boundary of what the present mathematics establishes. They are useful design information for future physical models.

A mathematical framework becomes more informative when its limits are known. The results collected here identify tempting shortcuts that fail: a coupling profile need not be selected by generic symmetry, a graph need not specify the physical metric, and a projected effective theory need not single out the sector realized in nature. Each statement has a particular scope, and the counterexamples help locate the missing assumptions.

The mature HSMT record includes explicit no-go theorems, non-selection results, and valid stops. Their logical distinction is important.

9.1 NI-0: monotone-coupling no-go

Consider

$$H(q) = H_* + \sum_e f_e(q_e) O_e, \qquad O_e \geq 0, \qquad f_e'(q_e) > 0. \tag{67}$$

Where the ground-state branch $E_0(q)$ is differentiable, Hellmann–Feynman gives

$$\frac{\partial E_0}{\partial q_e} = f_e'(q_e) \langle O_e \rangle_0 \geq 0. \tag{68}$$

No-Go Result 9.1 (NI-0: monotone positive-coupling no-go)**.** *Within the class* (67)*, ordinary q-independent interactions combined with monotone positive-semidefinite q-dependent couplings cannot by themselves generate the required interior restoring minimum along q.*

This is class-specific. It does not prove that no stabilization mechanism can exist after genuinely new structural input is added.

9.2 NI-1F: Gaussian measured-pullback monotonicity

For finite positive-semidefinite operators A and B under the frozen Gaussian measured-pullback construction, the associated real-positive functional $F(q)$ obeys

$$F'(q) \leq 0, \tag{69}$$

with equality only in the coincident-pullback case.

No-Go Result 9.2 (NI-1F)**.** *The frozen Gaussian measured-pullback response is monotone in q in the declared finite positive-semidefinite class and therefore does not generate an interior canonical vacuum selector through a sign change of $F'(q)$.*

The frozen NI-1 status is partial-positive structural / negative canonical-vacuum / new input required.

9.3 S+C minimal nonfundamental benchmark

The S+C family is retained explicitly as a *minimal nonfundamental benchmark*. An S move is a split/merge operation with

$$\Delta V = +1, \qquad \Delta E = +1, \tag{70}$$

while preserving β_1. A C move is a local cycle activation/deactivation with

$$\Delta V = 0, \qquad \Delta E = \pm 1, \tag{71}$$

and changes β_1. The firewall is

$$r \neq N, \qquad \chi = \kappa_C/\kappa_S \text{ unfixed.} \tag{72}$$

The family diagnoses refinement and cycle structure; it does not select a fundamental microscopic law.

9.4 B3, B5, and B7

The post-PI0 branches close as non-selection results:

- B3: the current premises do not select the proposed $V_{\mathrm{gr-rel}}$ bridge;
- B5: the current premises do not derive a distinguished seed or absolute grade origin;
- B7: the current premises do not select the candidate $W(\rho)$ law.

Each may define a valid model after new input; none is internally selected by the frozen core.

9.5 Transport non-selection

Locality, many-body dephasing, or clustering alone do not imply diffusion. A local unitary system may be ballistic, diffusive, anomalous, or localized depending on conserved quantities and current correlations. A genuine diffusive claim requires, in particular, absence of a nonzero ballistic/Drude component and suitable Green–Kubo integrability, schematically

$$D \propto \int_0^\infty \langle J(\tau)J(0)\rangle_c \, \mathrm{d}\tau \tag{73}$$

after the appropriate conserved-mode projections and normalization.

9.6 Positivity counterexample

The SRB audit distinguishes positive-semidefinite Gram structure, elementwise positivity, and strict positive overlap. These notions are not interchangeable. A frozen S^1 Fourier construction supplies a counterexample to the superseded stronger universal positivity claim. The exact Fourier formula is a source-fidelity item for the appendix; the scientific status is already frozen as a counterexample/no-go.

9.7 OMP-3F valid stop

Boundary Result 9.3 (OMP-3F)**.** *Within the analyzed packet/frame construction, the available premises do not establish either completeness or incompleteness of the candidate packet frame. The branch closes at a valid stop.*

9.8 BEGC no-smuggling comparison

The frozen BEGC-0 principle states:

> "At every physically admissible response level, externally relevant relations are generated from robustly controlled-complexity exposed states by a universal regular response law of robustly controlled nomological complexity. Independent pair-indexed response data are not fundamental inputs."

For

$$W_{ab} = F_\theta(z_a, z_b), \qquad \dim z = r, \qquad \dim \theta = p, \tag{74}$$

with r, p independent of n, the frozen finite-resolution comparison is

$$K_{\text{ext}}^{(\epsilon)} = O\left(nr \log \frac{1}{\epsilon} + p \log \frac{1}{\epsilon}\right) \tag{75}$$

versus generic pairwise data complexity

$$\Theta\left(n^2 \log \frac{1}{\epsilon}\right). \tag{76}$$

Thus the hierarchical family occupies positive codimension in the generic pair-data family for sufficiently large n. This is a conditional mathematical no-smuggling/compressibility result, not a physical dimension or selector theorem.

9.9 PB-0: physical projector principled stop

The frozen status is:

> **PB-0 = CLOSED / CURRENT CORE ANALYZES BUT DOES NOT IDENTIFY PHYSICAL P/Q / EMPIRICAL OR MICROSCOPIC BRIDGE REQUIRED.**

The exact frozen statements include:

> "The current core analyzes but does not identify a physical P/Q decomposition."
>
> "Physical-sector/projector identification is frozen as empirical or new microscopic bridge required."
>
> "Feshbach reduction is conditional dynamics given a decomposition, not a selector."
>
> "No further internal projector-selection search is licensed under the present core."

"Multi-contact response strengthens reconstruction and supplies cross-contact falsification but does not derive the grade decomposition."

"No concrete already-existing HSMT object operationally identifies a physical projector/ contact system rather than merely analyzing a chosen mathematical P."

9.10 PB0-N1 through PB0-N7

The following are reproduced verbatim from the frozen record.

Boundary Result 9.4 (PB0-N1)**.** *THE NUMERICAL VALUE OF c IS NOT AN INVARIANT GRADE SELECTOR. ONLY DIMENSIONLESS PROPAGATION RELATIONS OR THE CHARACTERISTIC / CAUSAL STRUCTURE HAVE AN INVARIANT CLAIM TO PHYSICAL SIGNIFICANCE.*

Boundary Result 9.5 (PB0-N2)**.** *A CHARACTERISTIC-PROPAGATION SELECTOR CAN IDENTIFY A GRADE ONLY TO THE EXTENT THAT THE GRADE-TO-CHARACTERISTIC MAP IS INJECTIVE OR OTHERWISE OPERATIONALLY DISCRIMINATING.*

Boundary Result 9.6 (PB0-N3)**.** *CHARACTERISTIC-PROPAGATION DATA MAY IDENTIFY A PHYSICAL GRADE REGIME WITHOUT FIXING AN ABSOLUTE GRADE ORIGIN.*

Boundary Result 9.7 (PB0-N4)**.** *A POSITIVE RELATIONAL / ELLIPTIC OPERATOR DOES NOT BY ITSELF DERIVE A LORENTZIAN CHARACTERISTIC CONE. PLACING THAT OPERATOR INTO A WAVE EQUATION IS AN ADDITIONAL DYNAMICAL CHOICE.*

Boundary Result 9.8 (PB0-N5)**.** *A POSTULATED GRADE-DEPENDENT SPEED c_N DOES NOT CONSTITUTE A PHYSICAL HSMT THEORY UNLESS THE GRADE DEPENDENCE, SCALE CALIBRATION, AND OBSERVABLE CONSEQUENCES ARE DERIVED INDEPENDENTLY.*

Boundary Result 9.9 (PB0-N6)**.** *A SPATIAL / RELATIONAL PRINCIPAL FORM ALONE CANNOT DETERMINE A PHYSICAL LIGHT SPEED. A RELATIVE NORMALIZATION BETWEEN TEMPORAL AND SPATIAL PRINCIPAL TERMS IS REQUIRED.*

Boundary Result 9.10 (PB0-N7)**.** *CAUSAL-CONE RECOVERY AND DIMENSION RECOVERY ARE DISTINCT PROBLEMS. DERIVING ONE DOES NOT BY ITSELF DERIVE THE OTHER.*

9.11 Global non-selection boundary

The present frozen premises do not derive:

(a) physical spacetime;

(b) observed physical dimension;

(c) operational physical time;

(d) Lorentzian causal structure;

(e) the uniquely realized physical P/Q decomposition;

(f) universal geometry across independently emergent sectors.

These are non-selection statements relative to the present premise set, not impossibility theorems against future extensions.

10 Foundational Closure and HSMT-FM0

After exploring many branches of the operator framework, a coherent picture emerges. The minimal architecture supports a substantial laboratory of exact reductions, spectral constructions, relational tests, and controlled examples. Its present derivational boundary is also identifiable. The checkpoint called HSMT-FM0 records that distinction: the current mathematical program has been organized to a defined boundary, while connecting it uniquely to the physical world calls for genuinely new input.

After organizing the constructive results, it is possible to state exactly where the present premise set stops. HSMT-FM0 is the resulting accounting checkpoint: it records the mathematical consequences already obtained and identifies external physical information that further interpretation would require.

The maturation program terminates internally at HSMT-FM0. The terminal statement is:

> "Internal derivation from the current premise set is complete to the presently identified logical boundary."

This explicitly does not mean that the physical theory is complete.

The mature five-stratum organization is:

F0 minimal primitives: $(\mathcal{H}, \{P_N\}, D)$ and, where used, DCP-0;

F1 internal mathematical consequences: operator control, projection, spectral structure, diagnostics, measurement, records, benchmarks;

F2 internal limits: no-go, non-selection, and valid-stop results;

F3 conditional physical bridges;

F4 external physical completion, presently absent.

Boundary Result 10.1 (HSMT-FM0)**.** *Given the currently frozen HSMT premises and completed audited branches, the internal open derivation ledger is empty to the presently identified boundary. Physical spacetime, physical dimension, Lorentzian causal structure, the uniquely realized physical sector, and multi-sector physical universality remain underived. Further physical closure therefore requires genuinely new information not already contained in the closed premise set.*

The status is

HSMT-FM0	FROZEN
Internal derivation	COMPLETE TO BOUNDARY
Physical theory	OPEN
Next legitimate activity	SYNTHESIS, EXPERIMENT, OR EXTERNAL-BRIDGE DISCOVERY

The selector role itself is identified as the MC-1 envelope, but no current result occupies it. PGCH, numerical c, a chosen P, an inserted c_N, a mode-count exponent, or a convenient Gaussian kernel do not fill the envelope merely by being mathematically useful.

11 Conditional Physical Bridges

A mathematical model reaches physics through a bridge: an independently justified reason to identify one of its quantities with a measurable property. For HSMT, such a bridge might connect a spectral response to a measured dispersion relation, a relational construction to operational distance, or an

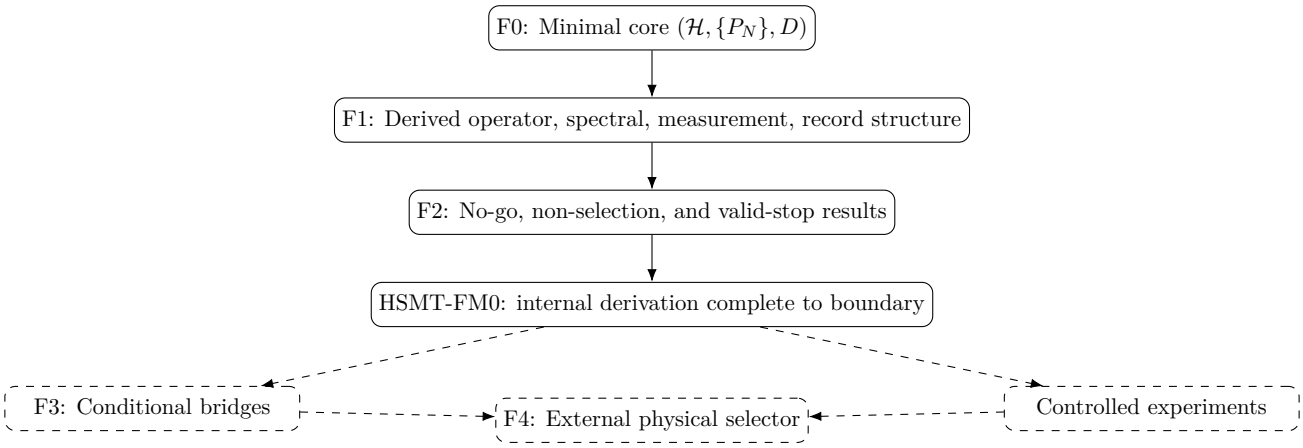


Figure 2: Logical architecture of mature HSMT. The transition from internal closure to physical selection remains open.

effective parameter to a calibrated clock. Stating these bridges explicitly turns vague possibilities into precise questions.

The architecture is designed to be connected to observations, but a connection must be made by an explicit bridge. Such a bridge may identify a sector with a measurable system, calibrate an evolution parameter, or give a physical meaning to a graph or spectral observable. Here we describe how those extensions can be made transparently and tested.

HSMT presently supports conditional statements of the form

$$\text{if bridge } B \text{ is independently justified, then consequence } C \text{ follows.} \tag{77}$$

The bridge itself is not selected by the current premise set.

Conditional Result 11.1 (Projected-sector bridge)**.** *If a particular mathematical sector P is independently identified as physically realized, then Theorem 5.1 supplies its exact reduced response under the stated operator hypotheses.*

Conditional Result 11.2 (Spectral/geometric bridge)**.** *If a declared spectral diagnostic is operationally shown to encode physical geometry, then its scale dependence may acquire physical geometric meaning.*

Conditional Result 11.3 (Dimensional bridge)**.** *If $Q_N(\ell)$ is connected to an operational physical observable satisfying the required scaling assumptions, then $d_N(\ell)$ may acquire physical dimensional meaning. Equation* (56) *alone does not provide that identification.*

Conditional Result 11.4 (Temporal bridge)**.** *If the mathematical evolution parameter τ is connected consistently to operational clocks and persistent record structure with the required normalization, an emergent physical-time interpretation may become available.*

Conditional Result 11.5 (Causal bridge)**.** *If a mathematical propagation structure is supplemented by an independently justified principle establishing operational signaling constraints and Lorentzian causal properties, it may support a physical causal interpretation.*

Conditional Result 11.6 (Multi-sector universality bridge)**.** *If cross-sector dynamics or empirical constraints force independently emergent sectors to share one effective geometry, their structures may support a common spacetime interpretation.*

PGCH is retained as a legitimate research hypothesis, not as an MC-1 selector. Likewise the historical chain (51) remains conditional.

11.1 Criterion for reopening derivation after FM0

A future foundational branch is legitimate only if it introduces genuinely new information. Qualifying inputs include:

(a) reproducible empirical discrimination among current realizations;

(b) an independently motivated microscopic law constraining grade, projector, generator, or bridge structure;

(c) an external mathematical principle not already encoded in the frozen core;

(d) an independently derived cross-theory identification that supplies an operational bridge without inserting the HSMT conclusion by hand.

Reparameterization, basis choice, preferred-grade declaration, fitted c_N, insertion of a wave equation, or renaming a count exponent as dimension do not qualify.

12 Computational and Hardware Benchmarks

Abstract operator equations can be made tangible in small, controllable systems. The HSMT benchmark program uses finite graphs and reversible oscillator realizations to produce exact spectra and measurable ratios. These models serve as test instruments: they let researchers check implementations and discriminate among proposed effective equations without conflating a deliberately engineered circuit with a fundamental law of nature.

Finite benchmarks turn the formalism into calculations that can be reproduced. Small graph Laplacians and reversible six-state oscillators supply exact spectra and distinguishable frequency ratios. A hardware realization tests whether specified dynamics can be implemented and measured; it can also reveal imperfections that an ideal calculation omits.

The benchmark program is explicitly separated into levels of evidence. Exact mathematics, numerical verification, emulator realization, SPICE implementation, physical hardware measurement, and physical-theory interpretation are distinct.

12.1 P3 exact benchmark

The three-site path is the smallest model that exhibits more than one nonzero collective mode. Its simple Laplacian can be diagonalized exactly, which makes it useful for testing the correspondence between matrix structure, mode frequencies, and experimental response.

The minimal path-graph Laplacian is

$$K = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}, \tag{78}$$

with

$$\operatorname{spec}(K) = \{0, 1, 3\}. \tag{79}$$

The positive spectrum is $\{1, 3\}$.

12.2 Six-real-state S and I realizations

A single mathematical graph can support more than one dynamical rule. The six-real-state constructions make that distinction observable by embedding alternative operators in coupled real variables. Their different frequency behavior demonstrates how a common state architecture need not determine a unique dynamics.

Let $u, v \in \mathbb{R}^3$ and $Y = (u, v)$. The S realization is

$$\dot{u} = gKv, \qquad \dot{v} = -gKu. \tag{80}$$

Hence

$$\ddot{u} = -g^2K^2u, \qquad \omega_S(\lambda) = g\lambda. \tag{81}$$

The I realization is

$$\dot{u} = gv, \qquad \dot{v} = -gKu, \tag{82}$$

so

$$\ddot{u} = -g^2Ku, \qquad \omega_I(\lambda) = g\sqrt{\lambda}. \tag{83}$$

Thus the spectral exponent is $\alpha_S = 1$ for S and $\alpha_I = 1/2$ for I.

For the P3 positive modes,

$$\frac{\omega_{S,2}}{\omega_{S,1}} = 3, \qquad \frac{\omega_{I,2}}{\omega_{I,1}} = \sqrt{3}. \tag{84}$$

12.3 P5 controlled discriminator

Moving from three sites to five creates a more demanding test: there are additional collective modes whose relative frequencies can distinguish models that look similar in simpler examples. This is an important bridge between a mathematical assertion and a practical laboratory discriminator.

The four positive P5 path eigenvalues are

$$\lambda_1 = \frac{3 - \sqrt{5}}{2}, \qquad \lambda_2 = \frac{5 - \sqrt{5}}{2}, \tag{85}$$

$$\lambda_3 = \frac{3 + \sqrt{5}}{2}, \qquad \lambda_4 = \frac{5 + \sqrt{5}}{2}. \tag{86}$$

Their approximate values are 0.381966011, 1.381966011, 2.618033989, and 3.618033989. The ratio targets are shown in Table 3.

Table 3: P5 frozen spectral-ratio targets relative to the lowest positive mode.

Ratio	R_{21}	R_{31}	R_{41}
S law, $\alpha = 1$	3.618033989	6.854101966	9.472135955
I law, $\alpha = 1/2$	1.902113033	2.618033989	3.077683537

For the general law $\omega_j = C\lambda_j^\alpha$, define

$$x_j = \ln \frac{\lambda_j}{\lambda_1}, \qquad y_j = \ln \frac{\widehat{\omega}_j}{\widehat{\omega}_1}. \tag{87}$$

A zero-intercept least-squares estimator is

$$\widehat{\alpha} = \frac{\sum_{j>1} x_j y_j}{\sum_{j>1} x_j^2}. \tag{88}$$

The overall calibration constant cancels. P5 is therefore stronger than P3 as a controlled discriminator because it provides three independent nontrivial ratio constraints rather than one.

12.4 EW-0 and implementation status

A physical device is never identical to its ideal equations. Components have finite bandwidth, switching transients, tolerance, and noise. The hardware plan therefore separates the ideal spectral prediction from the model-specific task of reproducing it with real or simulated electronics.

EW-0 is the frozen event-window synchronization protocol. Reconfiguration events are timestamped, settling windows are excluded or separately characterized, and S/I analysis windows use matched timing conventions so that switching transients are not mistaken for stationary spectral-law differences.

The intended mixed-model stack is:

- OPAx197 vendor operational-amplifier model;
- ADG333A vendor/ADI switch model;
- ADG1412 vendor switch model;
- datasheet-bounded behavioral LM319 comparator model.

The hardware checkpoint chain is

$$\text{H0-PASS} \to \text{HW-0} \to \text{POSTDR0-H2C-PARTIAL} \to \text{XR-0}. \tag{89}$$

XR-0 means SPICE execution-ready. The full vendor/behavioral transient remains unexecuted in the frozen record.

12.5 Evidence ladder

It is helpful to arrange evidence according to what each test can actually establish. Algebra establishes what a defined model predicts; a simulation checks an implementation; a bench experiment checks a physical device; and independent observations test whether the modeled behavior occurs in nature without being imposed by the experimenter.

The current status is summarized in Table 4.

The hardware program is therefore a controlled test of whether finite analog/emulator systems can realize and discriminate declared operator laws. It is not presently a direct experimental test of the physical selector problem.

13 Falsifiability and Empirical Access

A hypothesis becomes scientifically useful when it risks being wrong in a well-specified way. HSMT can generate exact conditional predictions for declared realizations. To test a physical claim, the parameters, measured quantities, and competing explanations must be fixed independently enough that failure cannot be repaired simply by renaming a sector or changing a coupling after observing the data.

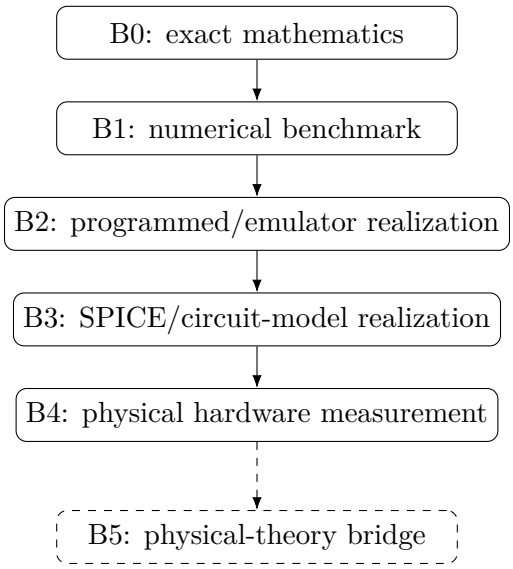


Figure 3: Benchmark evidence ladder. Even perfect implementation agreement at B4 does not by itself supply the B5 physical selector.

Table 4: Benchmark and hardware status.

Object	**Evidence level**	**Status**
Application A0	Numerical inverse benchmark	Completed
P3	Exact benchmark	Completed mathematically
P5	Exact controlled discriminator	Completed mathematically
Six-state S/I system	Programmed/emulator construction	Defined
EW-0	Measurement protocol	Frozen
Hardware package	Implementation specification	Frozen
XR-0	SPICE readiness	Execution-ready
Vendor/behavioral transient	Circuit-model execution	Not executed
Physical hardware measurement	Experimental	Not established
Physical-selector test	Physical-theory bridge	Open

For a framework intended to inform physics, the empirical question must be stated at the level of an observable. We distinguish verification of a programmable realization from evidence that a natural system chooses its own dynamics. This section sets out how controlled comparisons could constrain, rather than merely illustrate, the theoretical options.

Because HSMT presently admits multiple mathematical realizations, falsifiability must be stated at the correct logical level. A realization-level prediction may fail without falsifying every possible realization of the abstract framework. Conversely, a universal structural consequence, if one is identified, may constrain the framework itself.

Legitimate empirical targets include:

- failure of a bridge-specific scaling relation;
- failure of an emulator or hardware implementation to reproduce an exact benchmark within predeclared uncertainty;
- observational exclusion of a proposed physical selector;
- failure of a dimensional bridge to reproduce an operational dimension measurement;
- inconsistency of a proposed causal bridge with relativistic signaling behavior;
- incompatibility of a proposed universality mechanism with observed common spacetime structure.

The benchmark hierarchy in Fig. 3 therefore distinguishes implementation falsification from ontological falsification. Successful realization of the P3/P5 laws validates the implementation of those laws; it does not by itself establish that either is fundamental in nature.

14 Candidate Selection Principle: Maximal Collective Relaxation

The graded architecture tells us how different levels can interact, but it does not automatically determine which interaction strengths are realized. MCRSP-1 examines a concrete possibility: when a network's couplings can adapt while their total strength is conserved, perhaps its persistent organization favors rapid collective equilibration. This idea gives a precise optimization problem whose consequences can be derived, and it creates an experimentally testable distinction between weights chosen by a designer and weights selected spontaneously by a physical process.

The graded framework permits multiple interaction structures. MCRSP-1 explores one concrete way a particular structure might be chosen: adaptive couplings, subject to a fixed resource, favor configurations whose slowest collective disturbance relaxes as quickly as possible. The established mathematical object is a constrained graph optimization; the physical proposal is that an independently characterized natural adaptation process actually realizes its optimizer.

14.1 Scientific status and relationship to prior optimization literature

The quantity used here has a familiar role in diffusion: the slowest nonzero mode controls how quickly a connected system loses long-lived differences. The mathematical literature explains how to optimize such relaxation subject to constraints. MCRSP-1 asks a separate physical question: is there a natural adaptation mechanism that actually leads to that optimum?

The original graded core does not contain a mechanism selecting its physical generator. We therefore study a *separate, defeasible candidate* rather than silently adding a new axiom to the frozen core. The maximal collective-relaxation selection principle (MCRSP-1) proposes that, in a natural system with

physically adaptive conductances, fixed connected graph topology, fixed total positive conductance, and a conserved collective mode, the persistent selected interaction pattern maximizes the slowest nonzero relaxation rate. The mathematical optimization of mixing rates for continuous-time graph Markov processes is established in the literature [1, 2]. The speculative proposition here is *physical spontaneous selection*; neither the general optimization objective nor the path-graph algebra is claimed as a new physical law or a novel graph-theory invention.

Hypothesis 14.1 (MCRSP-1: conditional physical-selection proposal)**.** *Within a clearly specified physical class of independently adaptive, connected, weighted networks with conserved total conductance W and uniform response rates, the stable selected weights maximize $\lambda_2(L(w))$, the smallest positive eigenvalue of the weighted graph Laplacian, over the physically accessible set of weights.*

The word *independently* matters: the adaptation equations, conservation rule, topology, and accessibility must follow from physical characterization rather than be retrofitted to the proposed optimum. MCRSP-1 is not assumed to apply to all closed quantum systems or to all HSMT sectors. In particular, relaxation is not a property of an arbitrary unitary evolution.

14.2 Variational formulation

For a path-shaped network, each link receives a positive conductance. Holding their sum fixed prevents trivial improvement by simply increasing every coupling at once. What remains is a genuine allocation problem: how should a finite interaction resource be distributed so that the slowest collective disturbance decays as rapidly as possible?

For the path P_n with $n \geq 2$ vertices and edge conductances $w_i > 0$, define

$$L(w) = \sum_{i=1}^{n-1} w_i (e_i - e_{i+1})(e_i - e_{i+1})^{\mathsf{T}}, \qquad \sum_{i=1}^{n-1} w_i = W > 0. \tag{90}$$

The constant vector is the unique zero eigenvector. For the formal diffusion equation $\dot{p} = -L(w)p$, λ_2 controls the slowest nonstationary exponential decay. This parameter t is the time of the *declared diffusion model*, not an HSMT-derived clock. The conditional selection problem is

$$\underset{w_i>0,\ \sum_i w_i = W}{\text{maximize}} \quad \lambda_2(L(w)). \tag{91}$$

Theorem 14.2 (Optimal path weights)**.** *For the weighted n-vertex path with $n \geq 2$ and fixed sum of positive edge weights W, the unique maximizer of λ_2 is*

$$w_i^* = \frac{6W}{n(n^2-1)} i(n-i), \qquad 1 \leq i \leq n-1, \tag{92}$$

with optimum

$$\lambda_2(L(w^*)) = \frac{12W}{n(n^2-1)}. \tag{93}$$

Proof. Let $x_i = i - (n+1)/2$. Then $x \perp \mathbf{1}$, $x_{i+1} - x_i = 1$, and $\sum_i x_i^2 = n(n^2-1)/12$. The Rayleigh principle yields

$$\lambda_2(L(w)) \leq \frac{x^{\mathsf{T}} L(w) x}{x^{\mathsf{T}} x} = \frac{W}{n(n^2-1)/12}. \tag{94}$$

Write $c = 6W/[n(n^2-1)]$ and set $w_i = ci(n-i)$, with $w_0 = w_n = 0$. Since $(Lx)_i = w_{i-1} - w_i$, direct subtraction gives $Lx = 2cx$. The strictly increasing vector x has exactly one sign change (allowing a

central zero); the standard oscillation ordering for a positive Jacobi/path Laplacian places it in the first nonconstant eigenspace. Thus $\lambda_2 = 2c$ and the bound is attained. If another choice attains the same bound, x minimizes the Rayleigh quotient on $\mathbf{1}^\perp$ and hence satisfies $L(w)x = \lambda_2 x$. The first-row equation fixes w_1, and subsequent rows recursively fix every w_i, recovering $w_i = ci(n-i)$. The maximizer is therefore unique. □

14.3 Exact finite spectra and an independent discriminator

An optimization hypothesis is useful only if it leads to outcomes that differ from plausible alternatives. The five-site calculation supplies such an outcome: the optimum has stronger central links than end links, unlike the uniform distribution. Its eigenvalue ratios can be checked independently of the overall rate scale.

For $n = 3$ and $W = 2$, the optimum is $w_1 = w_2 = 1$, with $\operatorname{spec} L = \{0, 1, 3\}$. For $n = 5$ and $W = 4$, one obtains

$$(w_1, w_2, w_3, w_4) = (0.8, 1.2, 1.2, 0.8), \qquad \operatorname{spec} L = \{0, 0.4, 1.2, 2.4, 4.0\}. \tag{95}$$

The positive eigenvalue ratios are $1 : 3 : 6 : 10$. By contrast, unit weights on all four edges give eigenvalues $2 - 2\cos(k\pi/5)$, $k = 0, \ldots, 4$, and a different positive spectrum. Such ratios are predictions of the *specified matrices*. If a programmable emulator is built with weights w^*, observing those ratios tests implementation, not spontaneous adaptation.

For the three-site family $w_1 = 1 + \varepsilon$, $w_2 = 1 - \varepsilon$, $|\varepsilon| < 1$, the positive gap is

$$\lambda_2 = 2 - \sqrt{1 + 3\varepsilon^2} = 1 - \frac{3}{2}\varepsilon^2 + O(\varepsilon^4). \tag{96}$$

Thus small asymmetric perturbations reduce the gap quadratically; this follows mathematically without establishing an evolution law driving ε to zero.

14.4 Residual freedom: optimizing a grade block is not selecting physics

The selected coupling matrix controls only one part of a more general dynamics. Additional internal degrees of freedom can evolve differently while leaving that matrix unchanged. The following example shows this limitation concretely and helps identify what a stronger physical-selection principle would eventually have to determine.

Proposition 14.3 (Internal-generator nonselection)**.** *Let L^* be the five-site optimizer above and let B be an otherwise free self-adjoint operator on $\mathbb{C}^2$. The family*

$$H_B = L^* \otimes I_2 + I_5 \otimes B \tag{97}$$

has the same selected grade Laplacian and the same $\lambda_2(L^*)$, but it need not have unitarily equivalent full generators.

Proof. Take $B_0 = 0$ and $B_1 = \mu\sigma_z$ with $\mu \neq 0$. The first operator has each L^* eigenvalue twice; the second has eigenvalues $\lambda_j \pm \mu$. For generic nonzero μ the multisets differ. Both operators are self-adjoint and generate norm-preserving one-parameter unitary groups. Their collective Laplacian is identical. □

This proposition blocks the inference that selecting a Laplacian determines HSMT's physical Hamiltonian, its internal fibres, its grade domain, or physical spacetime. Extending MCRSP-1 to a general noncommutative generator would require an additional physically justified objective. No such extension has been derived.

14.5 Experimental test and falsification protocol

The decisive physical test would not be a circuit programmed to display the predicted ratio. It would be a system whose adaptation law is characterized independently and whose interactions evolve, without being forced toward the answer, to a measurable stable distribution. The distinction determines whether an experiment tests the optimization mathematics or the proposed natural selection mechanism.

Three noninterchangeable tests. *Mathematical optimization* verifies the extremizer of a declared objective; *engineered implementation* measures a system programmed with the proposed weights; *spontaneous physical selection* asks whether independently determined microscopic adaptation dynamics converge to those weights without being programmed to do so. Only the third test supplies direct evidence for the conjectured natural selection mechanism; even success there does not identify a fundamental HSMT law.

A genuine selection test must use a physical system whose adaptive rule and fixed-resource condition are independently characterized, rather than a network whose weights are manually programmed to match Eq. (92). Pre-specify the graph, equal node-response factors, conservation precision, observation horizon, admissible weight range, measurement uncertainties, and a quantitative criterion for convergence to a persistent configuration. For a qualifying five-site system with $W = 4$, MCRSP-1 predicts $w = (0.8, 1.2, 1.2, 0.8)$ up to the predeclared tolerances, with independently measurable relaxation spectrum as a cross-check. A reproducibly stable alternative pattern while the stipulated premises hold would reject MCRSP-1 *for that domain*. An engineered circuit reproducing Eq. (95) does not test spontaneous physical selection. No independent experiment of this kind is reported in the present manuscript.

14.6 Connection to the physical-selection ledger

The proposal has three distinct evidential layers: (i) a proven restricted graph optimization; (ii) an unverified hypothesis that certain natural adaptive networks spontaneously find its optimizer; (iii) an unestablished correspondence between such a network and HSMT's underlying dynamics. A positive result at (ii) would not prove (iii). In particular, MCRSP-1 does not derive $N = 4$, a Gaussian width, a physical arrow of time, Einstein gravity, Standard Model degrees of freedom, or a unique realized sector. The foundational selector remains open.

15 The Open Physical Selector Problem

HSMT supplies a mathematical language for possible levels of organization and a variety of methods for describing their effective behavior. What would make one of these possibilities the physical world? A satisfactory answer must identify the realized dynamics, connect its observables to measurement, and explain the origin of spacetime behavior rather than merely borrowing its terminology.

The preceding mathematics produces structured possibilities, and MCRSP-1 selects some interactions in a restricted family. To connect these possibilities to the world, HSMT still needs an independently warranted correspondence between the formal quantities and physical systems. This section identifies the particular pieces of that correspondence that remain open.

The next questions specify the observations and physical principles needed to connect this mathematical program with the world we measure.

Open Problem 15.1 (Physical selector). Which mathematical realization corresponds to the physical world?

Open Problem 15.2 (Physical spacetime bridge). What operational principle gives the abstract graded operator structure physical spacetime meaning?

Open Problem 15.3 (Physical dimension). What licenses identifying one mathematical dimensional diagnostic with observed physical dimension?

Open Problem 15.4 (Causal bridge). What establishes operational Lorentzian causal structure rather than only mathematical propagation?

Open Problem 15.5 (Multi-sector universality). Why should independently emergent sectors share one physical metric, dimensionality, causal cone, or spacetime realization?

Additional open items include the physical generator, the unique physical origin of the Gaussian kernel where not model-assumed, a possible grade-dependent width/separation law, packet-frame completeness, and empirical selection among admissible realizations.

15.1 Concise physical-completion ledger

For clarity to readers encountering HSMT for the first time, the missing physical identifications can be organized into four groups:

(i) **Selection:** an independently justified physical generator and realized P/Q sector, including a physical origin of coupling profiles;

(ii) **Spacetime:** operational time calibration, dimension, Lorentzian causal structure, and agreement across independently emergent sectors;

(iii) **Measurement:** a physically calibrated localization/POVM bridge, with packet-frame completeness established where needed;

(iv) **Discrimination:** empirical data that identify a physical realization rather than simply verifying an engineered emulator.

The complete itemized ledger appears in the appendix. No item in this summary is considered solved by the candidate principle MCRSP-1.

The MC-1 role is therefore an *envelope*: the function a genuine selector must perform has been identified, but no validated object occupies it. The conditional MCRSP-1 proposal in Section 14 optimizes a restricted collective Laplacian and demonstrably does not select the full generator.

16 Discussion

The most distinctive feature of HSMT is the attempt to place several questions usually treated separately inside one organized framework: how systems divide into effective levels, how those levels exchange influence, how observables depend on coarse-graining, and how one might eventually decide which effective description nature realizes. Its mathematical components are familiar in isolation; the intended contribution is their structured use in a common investigation of hierarchy and emergence.

The organizing contribution of HSMT is a common mathematical language for a hierarchy of sectors and the effective descriptions obtained from them. The grades allow one to state which degrees of freedom belong together; intergrade operators allow them to interact; the reduction identities explain how an apparently autonomous sector inherits corrections and memory from its surroundings. Spectral and scaling tools then turn those effective operators into questions about observable response. Relational

benchmarks demonstrate that these ideas can be calculated with exact matrices and reproducible dynamical systems.

This organization makes several avenues of investigation concrete. For example, one can ask when coupling to other grades is weak enough for a sector to behave approximately independently, or when memory kernels expose otherwise hidden degrees of freedom. One can compare different notions of effective dimension using declared diagnostics, rather than treating a dimensional label as a physical result. One can define and test finite network dynamics, including P3 and P5 spectral discriminators, with explicit scope for measurement error and hardware nonidealities. These are productive uses of the framework without requiring a premature interpretation as fundamental spacetime physics.

The search for a selector is equally constructive when posed carefully. MCRSP-1 turns a broad question about preferred interactions into a specific variational problem, yielding the nonuniform path weights and spectral predictions derived in Section 14. Its separate residual-generator counterexample shows why selecting couplings is not the same as selecting all physical dynamics. That is an informative outcome: it suggests that a future physical theory may need distinct principles for collective structure, internal dynamics, and operational interpretation.

The limitations can therefore be summarized in one place. The current premises do not derive a uniquely realized sector, a physical time calibration, Lorentzian causality, observed dimension, Standard Model dynamics, or the fundamental physical Hamiltonian. The no-go and valid-stop results specify where extra input is needed, not that the entire program is impossible. Ideas proposed during HSMT's development concerning cosmological redshift, CMB interpretations, and dimensionally layered physical behavior remain exploratory motivations and are not presented here as predictions.

The next advance should be judged by independent information: a microscopic physical mechanism, a controlled observable correspondence, or a discriminating experiment. The goal is not merely to find another mathematically admissible realization but to discover which aspects of the graded framework, if any, nature actually uses.

17 Conclusion

HSMT begins with a simple picture: what we observe may be one effective level of a more intricate organization, and the mathematics of those levels can be studied before their full physical interpretation is known. This paper has developed that picture through explicit graded operators, exact projection methods, spectral and measurement tools, and concrete finite models. Each offers a way to turn the language of shells into a problem with precise definitions and verifiable calculations.

Hierarchical Shell Manifold Theory offers an operator-based approach to hierarchical organization and emergent effective descriptions. Its core partitions a common Hilbert space into grades, permits interactions among them, and provides exact methods for studying reduced sectors. Self-adjoint realizations, Schur–Feshbach identities, memory kernels, spectral inverse problems, scale-dependent diagnostics, measurement constructions, and finite relational benchmarks form the mathematical substance of the program.

The MCRSP-1 proposal provides a particularly transparent illustration of how a selection question can become calculable. For a positive weighted path with fixed total conductance, its spectral-gap criterion gives a unique optimal interaction profile and distinctive finite spectra. The optimization theorem is rigorous within this model; spontaneous physical selection of that profile and its extension to the full HSMT generator remain hypotheses requiring independent evidence.

The central opportunity is now clear. HSMT supplies a structured setting in which to ask how effective sectors interact, persist, and acquire observable behavior. To turn that setting into an established

theory of nature requires an independent physical selector and operational bridges to space, time, and measurable phenomena. Presenting the constructive machinery alongside that precise research challenge is the purpose of this paper.

A Notation, Status Vocabulary, and Closure Terminology

Symbol/term	Meaning
$\mathcal{H}$	Hilbert-state arena.
P_N	Orthogonal grade projector.
N	Grade label; not physical dimension by definition.
D	Reference operator.
H	Generic self-adjoint dynamical/spectral operator when a specific evolution realization is used.
P, Q	Chosen effective/complement projections, $Q = I - P$.
τ	Mathematical evolution parameter; not automatically physical time.
$Q_N(\ell)$	Declared scale-dependent diagnostic associated with grade N.
$d_N(\ell)$	Local logarithmic scaling exponent of Q_N.
$d_{\mathrm{geom}}, d_{\mathrm{res}}, d_{\mathrm{spec}}, d_{\mathrm{dyn}}$	Distinct dimension notions; equality not assumed.
β_Ω	Historical inverse-square branch exponent.
β_{fr}	WKB frame-density exponent.
ν_B	Boundary/Bessel index in the conditional inverse-square branch.
HSMT-FM0	Terminal internal checkpoint: current-premise derivation complete to boundary, physical completion open.

Claim-status and closure terminology are defined in Section 10 and the opening methodological section. The distinction between no-go, non-selection, and valid stop must be maintained in any later extension.

B Operator Domains and Self-Adjointness Details

Theorem 4.1 is the principal operator-domain result required for the bounded-hopping constructions used in the main text. More generally, if a symmetric perturbation V is D_0-bounded with relative bound $a < 1$, i.e.

$$\|V\psi\| \le a\|D_0\psi\| + b\|\psi\|, \qquad \psi \in \mathrm{Dom}(D_0), \tag{98}$$

then Kato–Rellich yields self-adjointness of $D_0 + V$ on $\mathrm{Dom}(D_0)$. Bounded hopping is the special case $a = 0$.

The theorem concerns the declared realization. It does not select D_0, V, the grade couplings, or a physical generator.

C Exact Feshbach Reduction and Memory Kernel

For completeness, the block operator $z - H$ is

$$z - H = \begin{pmatrix} z - H_{PP} & -H_{PQ} \\ -H_{QP} & z - H_{QQ} \end{pmatrix}. \tag{99}$$

Eliminating the Q block by block Gaussian elimination gives the Schur complement (25) and projected resolvent (26). The time-domain kernel (34) follows by solving the complementary equation and substituting (31) into the P equation.

The exact initial-Q source in (33) is important: an initially occupied eliminated sector cannot generally be represented only by a homogeneous memory convolution.

D Spectral Inversion and Conditional Asymptotics

D.1 Stieltjes consistency checks

For a finite positive measure μ with convention (38), $m(z)$ is analytic on $\mathbb{C} \setminus \operatorname{supp} \mu$, satisfies $\operatorname{Im} m(z) < 0$ for $\operatorname{Im} z > 0$ unless μ is trivial, and obeys

$$m(z) = \frac{\mu(\mathbb{R})}{z} + O(z^{-2}) \tag{100}$$

when the required moments exist. For a normalized contact measure, $zm(z) \to 1$.

D.2 A0 noise response

The approximately linear transfer from multiplicative complex noise in G to relative error in Σ over the tested range is empirical to the frozen benchmark and should not be extrapolated without a separate conditioning analysis.

D.3 Historical chain

Equation (51) is retained exactly at conditional status. If the original source equations connecting A, M, μ, Σ, p, and E are recovered, they should be inserted here. In their absence the chain is not to be reverse-engineered into a new derivation during synthesis.

E Dimensional Diagnostics, SRB Counts, and MS-C0

For any positive differentiable $Q(\ell)$,

$$\frac{\mathrm{d} \ln Q}{\mathrm{d} \ln \ell} = \frac{\ell Q'(\ell)}{Q(\ell)}. \tag{101}$$

If $Q(\ell) = C\ell^d$, this equals d. If $Q(M) \sim CM^k$, the same identity gives a count exponent tending to k. Neither identity supplies physical dimensional meaning without an operational bridge.

The frozen SRB record also includes the onset datum $q_I(4) = 2$, the qualified Clifford-type identity for Δ_m, and the distinction between positive Gram structure, elementwise positivity, and strict overlap. Exact source formulas for q_I and the S^1 Fourier counterexample should be restored if available in the archival record.

F No-Go and Boundary Ledger

Item	Status	Frozen content
NI-0	No-go	Monotone positive-semidefinite q-couplings do not generate the required interior restoring minimum in the declared class.
NI-1F	Theorem/no-go	Gaussian measured-pullback functional is monotone; no canonical interior selector in the declared class.
B3	Non-selection	$V_{\mathrm{gr-rel}}$ not selected.
B5	Non-selection	Seed/absolute grade origin not derived.
B7	Non-selection	$W(\rho)$ not selected.
Transport	No-go against overreach	Locality/dephasing/clustering alone do not imply diffusion.
S^1 positivity	Counterexample	Positive Gram, elementwise positivity, and strict overlap are not equivalent.
OMP-3F	Valid stop	Packet-frame completeness remains undetermined.
MS-C0	No-go against conflation	Mode-count exponent is not physical dimension without an operational bridge.
BEGC	Conditional mathematical result	Exposed-state/universal-law family has lower finite-resolution complexity than generic pairwise data; not a physical selector.
PB-0	Principled non-selection	Chosen P/Q can be analyzed; physical P/Q is not selected.
PB0-N1–N7	Boundary sequence	Characteristic, grade-origin, causality, normalization, and dimension shortcuts excluded.
MC-1	Open envelope	Selector role identified; no occupant.
HSMT-FM0	Terminal internal boundary	Current-premise internal derivation complete to boundary; physical completion open.

G POVM, Packets, and Records

The publication-level summary retained from the frozen PC/OMP sequence is:

1. canonical POVM relative to the declared internal-fibre construction;
2. informational completeness on the internal fibre;
3. no canonical localized packet selected by the current internal structure;
4. packet-frame completeness unresolved at OMP-3F;
5. Q_N is a scalar diagnostic only;
6. RS-0 record persistence under (64) with the appropriate operator-domain conditions.

If the exact canonical POVM formula, PC wording, and OMP-3F source statement are recovered, they should replace this conservative archival summary in a subsequent publication revision.

H P3, P5, S+C, and Spectral-Law Benchmarks

H.1 P3 characteristic polynomial

For (78), direct evaluation gives

$$\det(K - \lambda I) = -\lambda(\lambda - 1)(\lambda - 3), \tag{102}$$

confirming (79).

H.2 General power-law discriminator

For $\omega = C\lambda^{\alpha}$,

$$\ln \frac{\omega_j}{\omega_1} = \alpha \ln \frac{\lambda_j}{\lambda_1}, \tag{103}$$

which motivates (88). The ratio test is calibration independent but not implementation-error independent.

H.3 S+C

S+C remains separate from P3/P5. The former is a minimal nonfundamental relational/refinement benchmark; the latter are controlled spectral dynamical benchmarks on a fixed finite operator. Neither supplies the physical selector.

I Hardware Architecture and Execution Status

The ideal state matrices corresponding to (80) and (82) are

$$A_S = \begin{pmatrix} 0 & gK \\ -gK & 0 \end{pmatrix}, \qquad A_I = \begin{pmatrix} 0 & gI \\ -gK & 0 \end{pmatrix}. \tag{104}$$

A physical or circuit realization has

$$A_{\text{real}} = A_{\text{target}} + \Delta A, \tag{105}$$

where ΔA collects finite gain/bandwidth, component tolerances, switch resistance/charge injection, comparator delay/hysteresis, slew rate, offsets, parasitics, supply limits, and switching transients.

A future result should predeclare ratio/exponent tolerances before inspecting the experimental outcome. A generic analysis statistic is

$$\epsilon_{\text{ratio}}^{(X)} = \left[\frac{1}{m} \sum_{j=2}^{m+1} \left(\frac{\widehat{R}_{j1} - R_{j1}^{(X)}}{R_{j1}^{(X)}} \right)^2 \right]^{1/2}, \tag{106}$$

with $X \in \{S, I\}$. This is an analysis proposal, not a new physical axiom; any exact frozen tolerance budget should supersede it.

The present archival status remains: execution-ready SPICE package, full vendor/behavioral transient unexecuted, physical bench measurement not established.

J Frozen Checkpoint Ledger

The mature checkpoint record is grouped by scientific role.

Checkpoint family	Scientific role
SRB-0–SRB-5	Structural/spectral audit: onset, mode counts, Δ_m identity, positivity qualifications, S^1 counterexample.
PCS / PCS-RP0	Finite-width feature map; valid execution / partial diagnostic failure; no RP continuation.
OMP sequence	Packet/frame program; OMP-3F valid stop.
PC-0–PC-4	Canonical POVM, internal-fibre IC, no canonical localized packet, Q_N scalar diagnostic.
FC-0	Minimal core $(\mathcal{H}, \{P_N\}, D)$ frozen.
PG-0	Physical generator not selected.
DCP-0	Norm-preserving closed evolution adopted.
PI-0	Evolution/time/causality firewall frozen.
B3/B5/B7	Post-PI0 non-selection results.
POSTFS0-SA1 / RS-0 / PSB1-B	Structural, record-persistence, and ring-exchange analogue branch.
H0-PASS / HW-0 / H2C-PARTIAL / XR-0 / EW-0	Exact benchmark through execution-ready hardware branch.
NI-0 / NI-1 / NI-1F	Monotone-coupling and Gaussian measured-pullback vacuum-selection no-go lineage.
S+C	Minimal nonfundamental refinement/cycle benchmark.
BEGC-0/1	No-smuggling / exposed-state universal-law complexity branch.
PB-0	Physical projector branch closed at principled non-selection.
PB0-N1–N7	Characteristic/causal boundary sequence.
HSMT-FM0	Terminal internal maturation checkpoint.

HSMT-FM0 is the terminal checkpoint of the present internal-development phase. It does not denote physical completion.

K Conditional Bridges, MC-1, and External Reopening

The MC-1 selector envelope is defined functionally: an occupant must discriminate among mathematically admissible realizations and connect the selected structure to operational observables through independently justified input. No current object occupies the envelope.

A future branch may reopen foundational derivation only through genuinely new empirical, microscopic, mathematical, or cross-theory information as described in Section 11. A conclusion-driven insertion does not count as a new bridge.

L Superseded Developmental Material and Provenance

The following classes of earlier developmental language are not live claims of the mature framework:

- literal ordinary concentric spatial-shell ontology as a foundational primitive;

- direct identification N = physical dimension;
- direct derivation of physical $3+1$ from the choice $N=4$;
- automatic elliptic-to-causal inference;
- unqualified equality $d_{\mathrm{geom}} = d_{\mathrm{res}} = d_{\mathrm{spec}} = d_{\mathrm{dyn}}$;
- treating coordinate reparameterization as dynamical selection;
- claiming locality/dephasing alone forces diffusion;
- positivity claims contradicted by the frozen S^1 counterexample;
- unverified screening, vacuum-selection, or cosmological claims promoted beyond their frozen status.

These items are retained here only to make the maturation boundary transparent.

Summary Status Tables

Table 8: Major positive results and their firewalls.

Result	**Status**	**Physical firewall**
Bounded-hopping self-adjointness	Theorem	Does not select physical generator
Exact Feshbach reduction	Exact theorem	Assumes P; does not select physical sector
Memory kernel	Exact derived result	Not an independent arrow of time
Scalar $G \to \Sigma$	Exact model identity	Does not select physical spectral measure
Semicircle A0	Analytic/numerical benchmark	Engine verification, not physical validation
WKB density map	Derived given W	Does not select W or identify frame index with grade
$Q_I, Q_H \sim M^3/6$	Structural asymptotic	Count exponent is not physical dimension
Canonical POVM / internal-fibre IC	Derived in frozen construction	Not physical position or spacetime localization
RS-0	Conservation condition	Not physical time or arrow
P3/P5	Exact controlled benchmarks	Not physical HSMT validation

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Table 9: Open physical ledger at HSMT-FM0.

Problem	**Status**
Physical selector	OPEN
Physical P/Q decomposition	OPEN
Physical spacetime bridge	OPEN
Physical dimension bridge	OPEN
Lorentzian causal bridge	OPEN
Operational physical-time bridge	OPEN
Multi-sector universality	OPEN
Physical generator	OPEN
Unique physical Gaussian-kernel origin	OPEN where not model-assumed
Grade-dependent width/separation law	OPEN
Packet-frame completeness	OPEN / VALID STOP
Empirical selection among admissible realizations	OPEN

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